

## 9.6 Expected mean squares

This section computes the expectations of sums of squares when the data are random, using only two moments, and their distributions under normality. Each mean square estimates  $\sigma^2$  plus a term that vanishes exactly when its hypothesis is true, while the residual mean square estimates  $\sigma^2$  alone. That is the structure of the  $F$  ratio, on which Part III builds.

### 9.6.1 Expected sums of squares for a decomposition

In this section  $E(\mathbf{Y}) = \boldsymbol{\mu}$  and  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ . We do not assume at first that  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ , because the effect of a wrong model on the mean squares is part of the story.

#### Theorem 9.6.1. Expected mean squares

Let  $\mathcal{V}_1, \dots, \mathcal{V}_k$  be an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ , with projections  $\mathbf{P}_i$  and dimensions  $r_i$ , and let  $E(\mathbf{Y}) = \boldsymbol{\mu}$ ,  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I}$ .

- (a)  $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 r_i + \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ , so the mean square of  $\mathcal{V}_i$  has expectation

$$E(\text{MS}_i) = \sigma^2 + \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{r_i}. \quad (9.6.1)$$

- (b)  $E(\text{SSE}) = \sigma^2(n - r) + \|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2$ , so  $E(s^2) = \sigma^2$  iff  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ .

- (c) If  $\boldsymbol{\mu} = \mathbf{X}\boldsymbol{\beta}$  with  $\mathbf{X} = [\mathbf{X}_0, \dots, \mathbf{X}_k]$  partitioned into terms, the partial sum of squares of  $\mathbf{X}_j$  has

$$E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_{(-j)}) = \sigma^2 d_j + \boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j,$$

with  $d_j = \text{rank}(\mathbf{M} - \mathbf{M}_{(-j)})$ , and the sequential sum of squares of  $\mathbf{X}_j$  has  $E \text{ SS}(\mathbf{X}_j \mid \mathbf{X}_0, \dots, \mathbf{X}_{j-1}) = \sigma^2 d'_j + \|(\mathbf{M}_j - \mathbf{M}_{j-1})\mathbf{X}\boldsymbol{\beta}\|^2$ , with  $d'_j = \text{rank}(\mathbf{M}_j - \mathbf{M}_{j-1})$ .

- (d) If moreover  $\mathbf{Y}$  is normal, then  $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ , and the sums of squares of different subspaces, including the error space, are uncorrelated.

*Proof.*

- (a) By Theorem 2.3.1 with  $\mathbf{A} = \mathbf{P}_i$  and  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ ,  $E(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = \sigma^2 \text{tr}(\mathbf{P}_i) + \boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu}$ . The trace is  $r_i$  (Proposition 6.3.4), and  $\boldsymbol{\mu}^\top \mathbf{P}_i \boldsymbol{\mu} = \boldsymbol{\mu}^\top \mathbf{P}_i^\top \mathbf{P}_i \boldsymbol{\mu} = \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ . Divide by  $r_i$ .
- (b) The same with  $\mathbf{A} = \mathbf{I} - \mathbf{M}$ , of trace  $n - r$ . The second term vanishes iff  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ .
- (c) For the partial sum of squares,  $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_l \boldsymbol{\beta}_l = \mathbf{0}$  for  $l \neq j$ , because  $\mathbf{X}_l \boldsymbol{\beta}_l$  lies in  $\mathcal{C}(\mathbf{X}_{(-j)})$  and is fixed by both projections. And  $(\mathbf{M} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j = \mathbf{X}_j \boldsymbol{\beta}_j - \mathbf{M}_{(-j)}\mathbf{X}_j \boldsymbol{\beta}_j = (\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j \boldsymbol{\beta}_j$ . Its squared length is  $\boldsymbol{\beta}_j^\top \mathbf{X}_j^\top (\mathbf{I} - \mathbf{M}_{(-j)}) \mathbf{X}_j \boldsymbol{\beta}_j$ , since  $\mathbf{I} - \mathbf{M}_{(-j)}$  is symmetric and idempotent. The sequential case is (a) for the chain decomposition (Definition 9.3.1).
- (d) By Theorem 4.2.3(b) with  $\boldsymbol{\Sigma} = \sigma^2 \mathbf{I}$ ,  $\text{Var}(\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}) = 2 \text{tr}\{(\sigma^2 \mathbf{P}_i)^2\} + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_i \boldsymbol{\mu} = 2\sigma^4 r_i + 4\sigma^2 \|\mathbf{P}_i \boldsymbol{\mu}\|^2$ . By part (d) of the same theorem the covariance of two forms is  $2\sigma^4 \text{tr}(\mathbf{P}_i \mathbf{P}_j) + 4\sigma^2 \boldsymbol{\mu}^\top \mathbf{P}_i \mathbf{P}_j \boldsymbol{\mu} = 0$  when  $\mathbf{P}_i \mathbf{P}_j = \mathbf{0}$ .

■

Parts (a) to (c) use nothing beyond second moments. They hold for skewed or heavy-tailed errors, provided the errors are uncorrelated with a common variance. Part (d) uses normality, and without it both the variances and the zero correlations fail in general ([Exercise \(C1\)](#)).

Part (c) confirms Proposition 9.3.1. For full-rank  $\mathbf{X}$  the term  $\|(\mathbf{I} - \mathbf{M}_{(-j)})\mathbf{X}_j\boldsymbol{\beta}_j\|^2$  is positive unless  $\boldsymbol{\beta}_j = \mathbf{0}$ , and it is large only if  $\mathbf{X}_j$  has a large part orthogonal to the other terms. A nearly collinear term has a small expected partial sum of squares even when its coefficients are substantial.

Under normality the expectations become full distributions.

### Corollary 9.6.2. Distributions of the mean squares

Let  $\mathbf{Y} \sim N_n(\boldsymbol{\mu}, \sigma^2 \mathbf{I})$  with  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ , and let  $\mathcal{V}_1, \dots, \mathcal{V}_k$  be an analysis of variance decomposition of  $\mathcal{C}(\mathbf{X})$ . Then the sums of squares  $\mathbf{Y}^\top \mathbf{P}_1 \mathbf{Y}, \dots, \mathbf{Y}^\top \mathbf{P}_k \mathbf{Y}$  and SSE are mutually independent,

$$\frac{\mathbf{Y}^\top \mathbf{P}_i \mathbf{Y}}{\sigma^2} \sim \chi^2(r_i, \gamma_i), \quad \gamma_i = \frac{\|\mathbf{P}_i \boldsymbol{\mu}\|^2}{\sigma^2}, \quad \frac{\text{SSE}}{\sigma^2} \sim \chi^2(n - r),$$

and  $F_i = \text{MS}_i / s^2 \sim F(r_i, n - r, \gamma_i)$ , which is central iff  $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$ . The same holds for a single projection  $\mathbf{P}$  onto any subspace of  $\mathcal{C}(\mathbf{X})$ , such as a partial sum of squares, paired with SSE.

*Proof.* The projections  $\mathbf{P}_1, \dots, \mathbf{P}_k, \mathbf{I} - \mathbf{M}$  are mutually orthogonal (Theorem 9.1.1(a)), so Theorem 4.4.4 gives independence and the noncentral chi-squared laws, with  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2 = 0$  for the error space. The ratio of two independent chi-squared variables, each divided by its degrees of freedom and the numerator noncentral, is the noncentral  $F$  (Definition 4.1.2). A single projection  $\mathbf{P}$  onto a subspace of  $\mathcal{C}(\mathbf{X})$  satisfies  $\mathbf{P}(\mathbf{I} - \mathbf{M}) = \mathbf{0}$ , and the same argument applies to the pair. ■

The residual part is the familiar Theorem 7.5.1; the rest applies the same argument to the other pieces.

## 9.6.2 The logic of the $F$ ratio

By [Eq. 9.6.1](#) every mean square estimates  $\sigma^2$  plus  $\|\mathbf{P}_i \boldsymbol{\mu}\|^2 / r_i$ , the squared size per dimension of the mean's component in  $\mathcal{V}_i$ , while under a correct model the residual mean square estimates  $\sigma^2$  alone. So the ratio

$$F_i = \frac{\text{MS}_i}{s^2}$$

compares two estimates of  $\sigma^2$  that agree in expectation exactly when  $\mathbf{P}_i \boldsymbol{\mu} = \mathbf{0}$ . This is the logic of the analysis of variance: *choose a denominator whose expectation equals that of the numerator when the hypothesis holds*. Independence gives the distribution exactly, and the noncentrality  $\gamma_i = \|\mathbf{P}_i \boldsymbol{\mu}\|^2 / \sigma^2$  governs the power. The probability that  $F_i$  exceeds any fixed critical value increases strictly with  $\gamma_i$  (Theorem 4.1.4). The mean of the ratio follows from independence and  $E(1/V) = 1/(\nu - 2)$  for  $V \sim \chi^2(\nu)$ ,  $\nu > 2$ :

$$E(F_i) = \frac{n - r}{n - r - 2} \cdot \frac{r_i + \gamma_i}{r_i}. \quad (9.6.2)$$

The principle is more general than the choice of  $s^2$ . With random effects the expected mean squares contain variance components, and the right denominator may be another mean square (Exercise (B1); Chapters 32 and 33, Proposition 32.1.1). Also, when the model is wrong, Theorem 9.6.1(b) shows that the denominator itself is inflated, by  $\|(\mathbf{I} - \mathbf{M})\boldsymbol{\mu}\|^2/(n - r)$  (compare Example (Residual sum of squares under a wrong mean)). Omitted terms inflate the denominator of every  $F$  ratio, which pushes the ratios towards zero, and they can also shift the numerators, because the omitted part of  $\boldsymbol{\mu}$  may have components in the  $\mathcal{V}_i$  (Exercise (B2)).

Chapter 11 derives the  $F$  test from the likelihood ratio principle (Theorem 11.4.1) and studies its power (Theorem 11.5.1).

### 9.6.3 Expected mean squares in an unbalanced layout

In the two-way layout of Section 9.3, Eq. 9.6.1 makes the hypotheses of Theorem 9.3.2 concrete. For the sequential sum of squares of  $A$  entered first,  $(\mathbf{P}_A - \mathbf{P}_0)\boldsymbol{\mu}$  has entry  $\bar{\mu}_i^w - \bar{\mu}^w$  on each of the  $n_{i\cdot}$  observations of row  $i$ , where  $\bar{\mu}^w = \sum_i n_{i\cdot} \bar{\mu}_i^w / n$ . So

$$\text{E MS}(A \mid \mu) = \sigma^2 + \frac{1}{a-1} \sum_i n_{i\cdot} (\bar{\mu}_i^w - \bar{\mu}^w)^2.$$

For the Type III sum of squares, the noncentrality is the sum of squares Eq. 9.2.1 evaluated at the true means, divided by  $\sigma^2$  (Exercise (B3)):

$$\gamma = \frac{(\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)^\top (\boldsymbol{\Lambda}^\top \mathbf{N}^{-1} \boldsymbol{\Lambda})^{-1} (\boldsymbol{\Lambda}^\top \boldsymbol{\mu}_c)}{\sigma^2},$$

where  $\boldsymbol{\mu}_c$  is the vector of  $ab$  cell means,  $\mathbf{N} = \text{diag}(n_{ij})$ , and the columns of  $\boldsymbol{\Lambda}$  express the  $a-1$  contrasts of unweighted row means. In the first, the design enters through the weights  $n_{ij}/n_{i\cdot}$ , which also define the hypothesis. In the second, the hypothesis does not depend on the design, which enters only through the precision matrix  $\mathbf{N}^{-1}$  of the cell means.

#### Example. Checking expected mean squares by simulation

Take the design of Example (Party, education and political self-placement):  $n = 944$  respondents in the nine cells of party by education, with the observed counts. Set the true cell means equal to the observed cell means, and  $\sigma^2 = 1.21$ . The script generates 40,000 data sets with normal errors and 40,000 with centred exponential errors of the same variance, which are strongly skewed. For each it computes the sequential decomposition (party, then education, then interaction), the residual, and the Type III sum of squares for party, and averages the mean squares:

| Source         | df  | $\gamma$ | E(MS)     | normal    | skewed    | Var(MS)  | simulated |
|----------------|-----|----------|-----------|-----------|-----------|----------|-----------|
| mean           | 1   | 14594.91 | 17661.050 | 17660.423 | 17661.259 | 85476.55 | 86357.17  |
| party          | 2   | 535.09   | 324.942   | 324.915   | 324.787   | 784.89   | 786.39    |
| education      | 2   | 35.75    | 22.840    | 22.836    | 22.891    | 53.81    | 53.13     |
| given<br>party |     |          |           |           |           |          |           |
| interaction    | 4   | 29.60    | 10.163    | 10.174    | 10.179    | 11.57    | 11.75     |
| residual       | 935 | 0.00     | 1.210     | 1.210     | 1.210     | 0.0031   | 0.0031    |

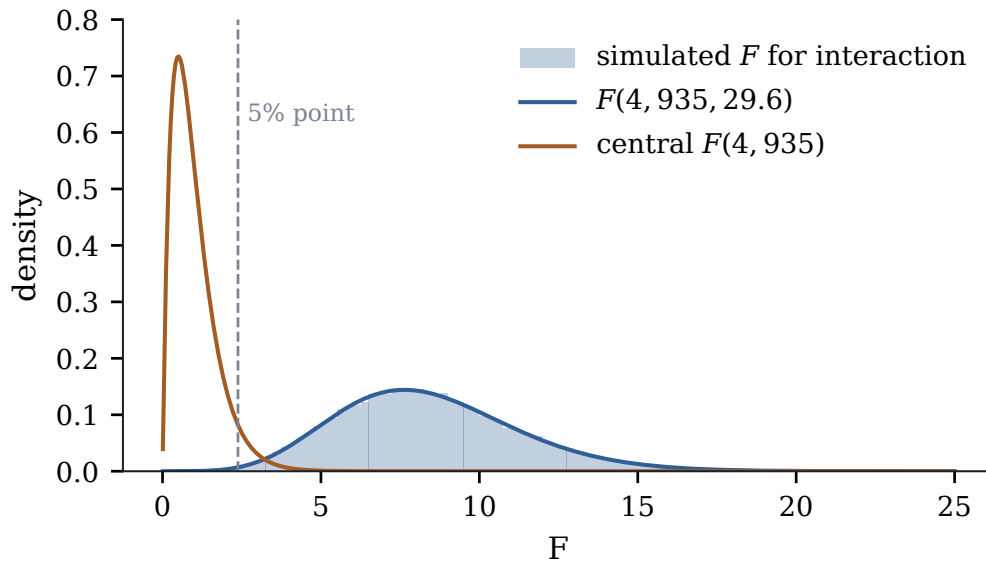


Figure 9.6.1: The  $F$  ratio for the interaction in [Example \(Checking expected mean squares by simulation\)](#) over 40,000 simulated data sets with normal errors, with the noncentral  $F(4, 935, 29.6)$  density of [Corollary 9.6.2](#) and the central null density  $F(4, 935)$  with its upper 5% point.

|          |   |        |         |         |         |        |        |
|----------|---|--------|---------|---------|---------|--------|--------|
| party,   | 2 | 537.53 | 326.414 | 326.362 | 326.259 | 788.46 | 789.47 |
| Type III |   |        |         |         |         |        |        |

The columns  $E(\text{MS})$  and  $\text{Var}(\text{MS})$  come from [Eq. 9.6.1](#) and [Theorem 9.6.1\(d\)](#), with  $\text{Var}(\text{MS}_i) = \sigma^4(2r_i + 4\gamma_i)/r_i^2$ ; the last column is the simulated variance under normal errors. The averages agree with the expectations under both error distributions, as parts (a) and (b) promise. The variances are another matter. Under skewed errors the variance of the residual mean square is 0.0124, 3.97 times the normal-theory value, because the variance of a quadratic form involves the fourth moment of the errors ([Theorem 2.3.3](#)). And the correlation between the interaction and residual mean squares, 0.008 under normal errors (zero in theory), is 0.020 under skewed errors, against a theoretical 0.025 from the third and fourth moments: the two mean squares are no longer independent.

For the interaction, the noncentrality  $\gamma = 29.60$  is large. By [Eq. 9.6.2](#), the  $F$  ratio has mean 8.417, and the simulated mean is 8.427. Its 5% critical value is 2.381, and the power is 0.9967 in theory and 0.9967 in the simulation. [Figure 9.6.1](#) shows the simulated distribution with the noncentral  $F$  density.

```
import numpy as np
import pandas as pd
import statsmodels.api as sm
d = sm.datasets.anes96.load_pandas().data
party = pd.cut(d["PID"], [-1, 1.5, 4.5, 6], labels=["Dem", "Ind", "Rep"])
educ = pd.cut(d["educ"], [0, 3.5, 5.5, 7], labels=["HS", "College", "Graduate"])
cell = party.cat.codes.to_numpy() * 3 + educ.cat.codes.to_numpy()
n = len(cell)

def basis(*blocks):
    Z = np.column_stack(blocks)
    U, s, _ = np.linalg.svd(Z, full_matrices=False)
```

```

    return U[:, s > 1e-10 * s[0]]

one = np.ones((n, 1))
A = np.eye(3)[party.cat.codes.to_numpy()]
B = np.eye(3)[educ.cat.codes.to_numpy()]
Cell = np.eye(9)[cell]
U0, UA, UAB, UC = basis(one), basis(one, A), basis(one, A, B), basis(Cell)

def complement(U_big, U_small):
    """Orthonormal basis of  $C(U\_small)$ -perp within  $C(U\_big)$ ."""
    R = U_big - U_small @ (U_small.T @ U_big)
    return basis(R)

pieces = {"mean": U0, "party": complement(UA, U0),
          "educ | party": complement(UAB, UA), "interaction": complement(UC, UAB)}
Ub = basis(one, B, np.column_stack([(A[:, [i]] - A[:, [0]]) * (B[:, [j]] - B[:,
[0]])
                                     for i in (1, 2) for j in (1, 2)]))
pieces["party (Type III)"] = complement(UC, Ub) # unweighted-means hypothesis

sigma = 1.1
mu = d.groupby(cell)["selfLR"].mean().to_numpy()[cell] # true means: observed cell
means
for name, U in pieces.items():
    r = U.shape[1]
    gamma = np.sum((U.T @ mu) ** 2) / sigma ** 2 # noncentrality  $\|P\mu\|^2 / \sigma^2$ 
    print(f"{name:17s} rank {r}  gamma = {gamma:9.2f}"
          f"  E(MS) = {sigma ** 2 * (1 + gamma / r):9.3f}")

```

[Run this code online](#)

```

rng = np.random.default_rng(99)
reps, chunk = 40_000, 4_000
dfe = n - 9
sums = {"normal": [], "skewed": []}
for kind in sums:
    out = []
    for _ in range(reps // chunk):
        if kind == "normal":
            E = sigma * rng.normal(size=(n, chunk))
        else:
            # centred exponential,
            # variance  $\sigma^2$ 
            E = sigma * (rng.exponential(size=(n, chunk)) - 1.0)
        Y = mu[:, None] + E
        ms = {name: np.sum((U.T @ Y) ** 2, axis=0) / U.shape[1] for name, U in
              pieces.items()}
        ms["residual"] = (np.sum(Y ** 2, axis=0) - np.sum((UC.T @ Y) ** 2, axis=0))
        / dfe
    out.append(pd.DataFrame(ms))
    sums[kind] = pd.concat(out, ignore_index=True)
print(pd.DataFrame({k: v.mean() for k, v in sums.items()})).round(3)

```

[Run this code online](#)

The simulation never forms an  $n \times n$  matrix: each sum of squares comes from the coordinates of  $\mathbf{Y}$  in an orthonormal basis of its subspace (Theorem 9.1.1(d)).

## 9.6.4 Exercises

### 9.6.4.1 A. Check your understanding

#### Exercise. A1

In simple regression with  $E(\mathbf{Y}) = \beta_0 \mathbf{1} + \beta_1 \mathbf{x}$ , show that the regression mean square has expectation  $\sigma^2 + \beta_1^2 S_{xx}$ , and that the  $F$  ratio of the corrected table has noncentrality  $\beta_1^2 S_{xx} / \sigma^2$  under normality.

#### Exercise. A2

In a one-way layout with  $g$  groups, group sizes  $n_i$  and means  $\mu_i$ , show that the between-groups mean square has expectation  $\sigma^2 + \sum_i n_i (\mu_i - \bar{\mu})^2 / (g - 1)$ , where  $\bar{\mu} = \sum_i n_i \mu_i / n$ . Compare with Example (The one-way layout).

### 9.6.4.2 B. Practice

#### Exercise. B1

In a balanced one-way layout with  $g$  groups of  $m$  observations, let  $Y_{ij} = \mu + a_i + \varepsilon_{ij}$ , where the  $a_i$  and  $\varepsilon_{ij}$  are uncorrelated random variables with mean zero and variances  $\sigma_a^2$  and  $\sigma^2$ . Show that  $\text{Cov}(\mathbf{Y}) = \sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top$ , with  $\mathbf{Z}$  the matrix of group indicators, and use Theorem 2.3.1 to show that

$$E(\text{MS}_{\text{between}}) = \sigma^2 + m\sigma_a^2, \quad E(\text{MS}_{\text{within}}) = \sigma^2.$$

Which ratio tests  $\sigma_a^2 = 0$ , and what unbiased estimator of  $\sigma_a^2$  do the two mean squares suggest?

*Solution.*  $\mathbf{Y} = \mu \mathbf{1} + \mathbf{Z}\mathbf{a} + \boldsymbol{\varepsilon}$ , so  $E(\mathbf{Y}) = \mu \mathbf{1}$  and  $\text{Cov}(\mathbf{Y}) = \mathbf{Z}(\sigma_a^2 \mathbf{I}_g) \mathbf{Z}^\top + \sigma^2 \mathbf{I}$  by Theorem 2.2.1. The between-groups projection is  $\mathbf{M} - \mathbf{P}_1$ , where  $\mathbf{M} = m^{-1} \mathbf{Z}\mathbf{Z}^\top$  averages within groups, and the within-groups projection is  $\mathbf{I} - \mathbf{M}$ . The mean  $\mu \mathbf{1}$  is annihilated by both. By Theorem 2.3.1,  $E(\mathbf{Y}^\top (\mathbf{M} - \mathbf{P}_1) \mathbf{Y}) = \text{tr}\{(\mathbf{M} - \mathbf{P}_1)(\sigma^2 \mathbf{I} + \sigma_a^2 \mathbf{Z}\mathbf{Z}^\top)\}$ . Now  $\mathbf{Z}\mathbf{Z}^\top = m\mathbf{M}$ , and  $(\mathbf{M} - \mathbf{P}_1)\mathbf{M} = \mathbf{M} - \mathbf{P}_1$ , so the trace is  $\sigma^2(g - 1) + \sigma_a^2 m(g - 1)$ . For the within-groups projection,  $(\mathbf{I} - \mathbf{M})\mathbf{Z} = \mathbf{0}$ , so the trace is  $\sigma^2(n - g)$ . Dividing by the degrees of freedom gives the two expectations. The ratio  $\text{MS}_{\text{between}}/\text{MS}_{\text{within}}$  has expectations that agree iff  $\sigma_a^2 = 0$ , and  $(\text{MS}_{\text{between}} - \text{MS}_{\text{within}})/m$  is unbiased for  $\sigma_a^2$  (though it can be negative).

#### Exercise. B2

Suppose  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta} + \mathbf{z}\delta$  but the model  $E(\mathbf{Y}) = \mathbf{X}\boldsymbol{\beta}$  is fitted, with  $\mathbf{1} \in \mathcal{C}(\mathbf{X})$ . Show that

$$E(\text{SSE}) = \sigma^2(n - r) + \delta^2 \|(\mathbf{I} - \mathbf{M})\mathbf{z}\|^2, \quad E(\text{SSR}) = \sigma^2(r - 1) + \|(\mathbf{M} - \mathbf{P}_1)(\mathbf{X}\boldsymbol{\beta} + \mathbf{z}\delta)\|^2.$$

Give an example in which  $\boldsymbol{\beta}$  has all slopes zero but the regression mean square is inflated.

#### Exercise. B3

In the setting of Theorem 9.2.4 with  $\mathbf{d} = \mathbf{0}$ , show that the expected sum of squares of the

hypothesis  $\mathbf{\Lambda}^\top \boldsymbol{\beta} = \mathbf{0}$  is  $\sigma^2 q + (\mathbf{\Lambda}^\top \boldsymbol{\beta})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \boldsymbol{\beta}$ , so the noncentrality is obtained by evaluating Eq. 9.2.1 at the true parameter.

*Solution.* By Theorem 9.2.4(b), the sum of squares is  $\mathbf{Y}^\top \mathbf{K} \mathbf{Y}$  with  $\mathbf{K} = \mathbf{M} \mathbf{T} (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{T}^\top \mathbf{M}$ , a projection of rank  $q$ , where  $\mathbf{T}$  is the  $n \times q$  matrix with  $\mathbf{\Lambda} = \mathbf{X}^\top \mathbf{T}$ . By Theorem 9.6.1(a) its expectation is  $\sigma^2 q + \boldsymbol{\beta}^\top \mathbf{X}^\top \mathbf{K} \mathbf{X} \boldsymbol{\beta}$ . Since  $\mathbf{T}^\top \mathbf{M} \mathbf{X} \boldsymbol{\beta} = \mathbf{T}^\top \mathbf{X} \boldsymbol{\beta} = \mathbf{\Lambda}^\top \boldsymbol{\beta}$ , the second term is  $(\mathbf{\Lambda}^\top \boldsymbol{\beta})^\top (\mathbf{\Lambda}^\top \mathbf{G} \mathbf{\Lambda})^{-1} \mathbf{\Lambda}^\top \boldsymbol{\beta}$ .

### 9.6.4.3 C. Going deeper

#### Exercise. C1

Let the errors be independent with mean zero, variance  $\sigma^2$  and fourth moment  $\mu_4 = E(\varepsilon_i^4)$ , and let  $\boldsymbol{\mu} \in \mathcal{C}(\mathbf{X})$ . Use Theorem 2.3.3 to show that

$$\text{Var}(\text{SSE}) = (\mu_4 - 3\sigma^4) \sum_i (1 - h_{ii})^2 + 2\sigma^4(n - r),$$

where  $h_{ii}$  are the leverages. For the centred exponential errors of Example (Checking expected mean squares by simulation),  $\mu_4 = 9\sigma^4$ . Show that  $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$  when all leverages are small, and compare with the simulated variance of the residual mean square.

*Solution.* With  $\mathbf{A} = \mathbf{I} - \mathbf{M}$  and  $\boldsymbol{\theta} = \boldsymbol{\mu}$ , the terms of Theorem 2.3.3 involving  $\boldsymbol{\theta}$  vanish because  $(\mathbf{I} - \mathbf{M})\boldsymbol{\mu} = \mathbf{0}$ . The diagonal of  $\mathbf{I} - \mathbf{M}$  is  $1 - h_{ii}$ , and  $\text{tr}\{(\mathbf{I} - \mathbf{M})^2\} = n - r$ , which gives the formula. If the leverages are small,  $\sum_i (1 - h_{ii})^2 \approx n - r$  (it equals  $n - 2r + \sum_i h_{ii}^2$ ), so  $\text{Var}(\text{SSE}) \approx (6 + 2)\sigma^4(n - r)$  and  $\text{Var}(s^2) \approx 8\sigma^4/(n - r)$ , four times the normal value  $2\sigma^4/(n - r)$ . This is the factor seen in the simulation.

#### Exercise. C2

For the Type I sum of squares of  $A$  entered first in a two-way layout, show that its noncentrality can be positive when  $\mu_{ij} = \beta_j$  depends on  $B$  only, and find the condition on the counts  $n_{ij}$  under which it is zero for all such  $\boldsymbol{\mu}$ . Relate your answer to Theorem 9.4.3.